

A SHARP SQUARE-ROOT ATLAS FOR MIDDLE ERDŐS–STRAUS STATES AND AN OBSTRUCTION TO FIXED GRADES

THE CLANKERS

ABSTRACT. For every prime $p \equiv 1 \pmod{8}$, an oriented first-half middle state has two odd cofactors $R, Q \equiv 3 \pmod{4}$. We prove the sharp inequality $4p \geq 3 \min(R, Q)^2 + 6 \min(R, Q) - 25$. On the six hard residue classes modulo 840 this improves to $4p \geq 3 \min(R, Q)^2 + 14 \min(R, Q) - 81$, with a complete equality classification and an attained example at $p = 1801$. The resulting $O(\sqrt{p})$ cutoff yields a disjoint, invertible two-chart atlas: a direct chart in the original square-divisor coordinates and a cofactor chart with $u \mid j^2$ and $4j - 1 \mid p + 4u$. Every coordinate map, inverse, ordering condition and exceptional tie is retained. We also prove unconditionally that no fixed prime-independent cutoff on the original exterior grade h and middle grade j can cover all hard primes, even though the constructed primes have explicit endpoint solutions. Two independent programs replay the finite atlas through 3000 and check a partial-factor order certificate. Universal Erdős–Straus occupancy is not proved: the remaining problem is to force at least one original E or M coefficient at every hard prime.

1. SCOPE AND ORIGINAL ARITHMETIC

The universal existence obligation is still outstanding. This note proves a sharp square-root cutoff for a *complete two-chart atlas of middle states*, not that this atlas is occupied at every prime. It also rules out a fixed, prime-independent cutoff on the original exterior grade h and middle grade j . Neither statement resolves the Erdős–Straus conjecture.

Let $p \equiv 1 \pmod{8}$ be prime. Throughout, divisors are positive divisors. Call p *hard* when

$$p \bmod 840 \in \{1, 121, 169, 289, 361, 529\}.$$

An original first-half middle state consists of

$$p/4 < a < p/2, \quad R = 4a - p, \quad u \in \mathbb{Z}_{>0}, \quad u \mid a^2, \quad R \mid u + a.$$

Write $a = \prod_l l^{e_l}$ and retain $\beta_l = v_l(u) - e_l \in [-e_l, e_l]$. Divisor complementation $u \mapsto a^2/u$ exchanges the last two ordered denominators. We choose $u < a$, recording the orientation bit; $u = a$ cannot be a middle state. With $g = \gcd(a, u)$ put

$$(h, r, s) = (g^2/u, u/g, a/g), \quad \lambda = (r + s)/R.$$

Then $a = hrs$, $u = hr^2$, $r < s$, $\gcd(r, s) = 1$, and

$$\frac{4}{p} = \frac{1}{a} + \frac{1}{phs\lambda} + \frac{1}{phr\lambda}. \tag{1}$$

The ordered inverse is $u = pa^2/(Ry - pa)$. These are the inherited Type-II coordinates [1, §2, Proposition 2.6]. All source exponents, the orientation bit and both complementary divisors are retained.

An original first-half exterior state has the same positive data $p/4 < a < p/2$, $R = 4a - p$, $u \mid a^2$ and the exterior gate $R \mid 4u + 1$. The same gcd normalization gives $a = hrs$, $u = hr^2$ and $\gcd(r, s) = 1$. Since

$$pr + s \equiv s(4hr^2 + 1) \equiv 0 \pmod{R},$$

the integer $\kappa = (pr + s)/R$ returns the ordered triple

$$(a, hsk, phr\kappa).$$

Indeed its last two reciprocals sum to $(pr + s)/(phr\kappa) = R/(pa)$, so the three reciprocals sum to $4/p$. These exterior coordinates are used only in the fixed-grade theorem below; they are not identified with the middle coordinates.

Set

$$j = hr\lambda, \quad Q = 4j - 1, \quad A = h\lambda^2.$$

The original equations give

$$QR = p + 4hr^2, \quad Qs = p\lambda + r, \quad Q + 1 < 2AR, \tag{2}$$

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$$\boxed{p = RQ - \frac{(Q+1)^2}{4A}}. \quad (3)$$

Both R, Q are $3 \pmod{4}$ and lie in $(0, p)$: indeed $hr^2 < a < p/2$, so $QR < 3p$ and $R \geq 3$. No coprimality between R and Q is required. The strict inequality in (2) is exactly $r < s$.

2. A SHARP HYPERBOLA BOUND

Theorem 2.1 (All $p \equiv 1 \pmod{8}$). *Let $t = \min(R, Q)$ in an original oriented middle state. Then*

$$\boxed{4p \geq 3t^2 + 6t - 25}. \quad (4)$$

Equivalently, putting $K_0(p) = \lfloor \sqrt{(p+7)/12} \rfloor$,

$$\min(R, Q) \leq 4K_0(p) - 1.$$

The constant is attained at $p = 41$.

Proof. First suppose $A \geq 2$. If $Q = t$, equation (3) immediately gives

$$p \geq t^2 - \frac{(t+1)^2}{4A} \geq \frac{7t^2 - 2t - 1}{8}.$$

If $R = t$, the function $tQ - (Q+1)^2/(4A)$ is strictly increasing on the actual interval $Q+1 < 2At$, so the same bound follows by setting $Q = t$. Its difference from $(3t^2 + 6t - 25)/4$ is $(t-7)^2/8$.

Now $A = 1$ implies $h = \lambda = 1$, $Q = 4r - 1$, $R = r + s$. The tie $R = Q$ would give $s = 3r - 1$ and $p = (2r - 1)(6r - 1)$; its only possible prime case $r = 1$ gives $p = 5$, outside the domain. If $Q = t < R$, then $R \geq t + 4$, and

$$p \geq \frac{3t^2 + 14t - 1}{4} > \frac{3t^2 + 6t - 25}{4}.$$

If $R = t < Q$, write $Q = t + 4k$. The original strict order yields $1 \leq k \leq (t-3)/4$, so $t \geq 7$, and

$$p = f_t(k) := \frac{3t^2 + (8k-2)t - (4k+1)^2}{4}. \quad (5)$$

This function is concave. At the first endpoint it equals $D(t) := (3t^2 + 6t - 25)/4$. At the last endpoint it is $t^2 - t - 1$, whose excess over $D(t)$ is $(t-3)(t-7)/4 \geq 0$. This proves (4). Finally $t \equiv 3 \pmod{4}$, so write $t = 4k - 1$. Inequality (4) is then exactly $p \geq 12k^2 - 7$, or $k^2 \leq (p+7)/12$. Thus it is equivalent to $k \leq \lfloor \sqrt{(p+7)/12} \rfloor = K_0(p)$ and hence to the displayed cutoff on t . At $p = 41$ it is attained by $(h, r, s, \lambda) = (1, 3, 4, 1)$ and by $(2, 1, 6, 1)$, both with $a = 12$ and $t = 7$. $\square$

Theorem 2.2 (Sharp hard-prime refinement). *Suppose additionally*

$$p \bmod 840 \in \{1, 121, 169, 289, 361, 529\}.$$

Then

$$\boxed{4p \geq 3t^2 + 14t - 81}. \quad (6)$$

Let

$$K(p) = \left\lfloor \frac{\sqrt{3p+73} - 2}{6} \right\rfloor, \quad B(p) = 4K(p) - 1.$$

Then $\min(R, Q) \leq B(p)$. Equality in (6) forces

$$h = \lambda = 1, \quad Q = R + 8, \quad s = 3r - 9, \quad p = 12r^2 - 40r + 9. \quad (7)$$

It is attained at $p = 1801$, $(h, r, s, \lambda) = (1, 14, 33, 1)$.

Proof. A hard prime is at least 1009; this also follows by checking the six possible progressions below that number. For $A \geq 2$, the preceding bound has excess $(t-7)(t-23)/8$ over the right side of (6) divided by four. This is nonnegative for $t \geq 23$. For $t \leq 23$, that latter expression is at most 457, strictly below p . Thus equality is impossible in this case.

For $A = 1$, the tie is already excluded. If $Q = t < R$, the bound in the previous proof exceeds the desired one by 20. In (5), the case $k = 1$ would give $p = 12(r-1)^2 - 7 \equiv 2 \pmod{3}$, impossible for a hard prime. Hence $k \geq 2$, which requires $t \geq 11$. Concavity reduces to $k = 2$ and $k = (t-3)/4$. The first is exactly (6); the last has excess $(t-7)(t-11)/4 \geq 0$. The possible equality at $t = 11$ gives $p = 109$, not hard. All remaining equality cases have $k = 2$, giving (7). For the cutoff formula, write the cofactor minimum as $t = 4m - 1$. Inequality (6) is exactly

$$p \geq 12m^2 + 8m - 23.$$

The positive root of $12X^2 + 8X - (p + 23)$ is $(\sqrt{3p + 73} - 2)/6$; therefore (6) is equivalent to $m \leq K(p)$ and to $t \leq B(p)$. For $p = 1801$, $a = 462$, $u = 196$, $R = 47$, $Q = 55$ and the ordered triple is $(462, 59433, 25214)$. Direct multiplication verifies (1). More generally $r = 14 + 210n$, $s = 3r - 9$ gives positive integral data with $\gcd(r, s) = 1$, $p = 12r^2 - 40r + 9 \equiv 121 \pmod{840}$ and equality in (6) for every $n \geq 0$. These data are instances of the hard-prime theorem exactly when the displayed value of p is prime; $n = 0$ gives $p = 1801$, whereas, for example, $n = 1$ gives $593161 = 19 \cdot 31219$. No infinitude of prime values of this quadratic is asserted. $\square$

The hard restriction cannot simply be replaced by $p \equiv 1 \pmod{24}$. At $p = 193$, $(h, r, s, \lambda) = (2, 2, 13, 1)$ has $R = Q = 15$ and violates (6). The valid general inequality (4) is retained there.

3. THE COMPLETE DISJOINT TWO-CHART ATLAS

Theorem 3.1 (Original-coordinate completeness). *For a hard prime, every middle state with its chosen $r < s$ orientation occurs in exactly one of the following charts.*

Direct chart:

$$\begin{aligned} 3 \leq R \leq B(p), \quad R \equiv 3 \pmod{4}, \quad a = (p + R)/4, \\ u \in \mathbb{Z}_{>0}, \quad u \mid a^2, \quad u < a, \quad R \mid u + a, \quad Q = (p + 4u)/R, \quad R \leq Q. \end{aligned} \quad (8)$$

Cofactor chart:

$$1 \leq j \leq K(p), \quad Q = 4j - 1, \quad u \in \mathbb{Z}_{>0}, \quad u \mid j^2, \quad Q \mid p + 4u, \quad R = (p + 4u)/Q > Q. \quad (9)$$

The inverse from (9) is

$$g = \gcd(j, u), \quad \boxed{h = g^2/u, \quad r = u/g, \quad \lambda = j/g, \quad s = R\lambda - r,} \quad (10)$$

$$a = Rj - u = hrs,$$

and then (1). The tie $R = Q$ belongs only to (8).

Proof. Equation (2) sends every original state with $Q < R$ to (9), since $j^2/u = h\lambda^2$ is integral. Theorem 2.2 puts its j in the stated range. A state with $R \leq Q$ goes to (8).

Conversely, data in the direct chart already have the original positive divisor and congruence conditions. To check the interval explicitly, put $K = K(p)$. A hard prime has $K \geq 2$, and the floor definition gives $p \geq 12K^2 + 8K - 23$. Hence

$$p - B(p) \geq 12K^2 + 4K - 22 > 0.$$

Thus $0 < R < p$ and $p/4 < a < p/2$. Moreover $p + 4u = 4(a + u) - R$, so the displayed Q is integral. The gcd normalization in the first section now reconstructs the unique positive (h, r, s, λ) and (1).

For data in the cofactor chart, the primewise valuation inequality $u \mid j^2$ proves the integrality of $h = g^2/u$ in (10), and gives $j = hr\lambda$, $u = hr^2$, $\gcd(r, \lambda) = 1$. The relation $RQ = p + 4u$ becomes $p = 4hr(R\lambda - r) - R$, proving $a = hrs$.

The range entails $p > 4j^2 - 2j$. Indeed, $j \leq K(p)$ gives $p \geq 12j^2 + 8j - 23$; the claimed strict inequality follows for $j \geq 2$, and for $j = 1$ from $p \geq 1009$. Since $u \leq j^2$,

$$jp > u(4j - 2)$$

implies $Rj > 2u$, hence $u < a$ and $s > r > 0$. Also

$$p(Q - 1) - 4u > 0 :$$

for $j = 1$ this is $2p - 4 > 0$, and for $j \geq 2$ the lower bound $p > 2j(2j - 1)$ makes it greater than $4j\{(2j - 1)^2 - j\} > 0$. Thus $R < p$. Both factors Q, R are $3 \pmod{4}$, so $p/4 < a < p/2$.

As $r \leq j < p$, a prime dividing r and R would divide p by $RQ = p + 4u$, which is impossible. Hence $\gcd(r, s) = \gcd(r, R\lambda) = 1$. The original divisor and gate relations are now explicit:

$$\frac{a^2}{u} = hs^2 \in \mathbb{Z}_{>0}, \quad a + u = Rj.$$

Also $\gcd(a, u) = hr$ and $\gcd(j, u) = hr$, since $\gcd(r, s) = \gcd(r, \lambda) = 1$. The original normalization at (a, u) therefore recovers exactly (10). Its inverse reads $j = hr\lambda$ and the same integer u . Within either chart, $R = 4a - p$, u , $Q = (p + 4u)/R$ and $j = (Q + 1)/4$ are fixed, and the gcd normalization is unique. The disjoint inequalities in (8)–(9) prove cross-chart uniqueness. Restore the recorded orientation by $u \mapsto a^2/u$ and swapping the last two denominators. $\square$

The congruence $u \mid j^2, 4j-1 \mid p+4u$ is derived here from the original Type-II coordinates. For historical placement, Elsholtz–Tao [1, §2, Proposition 2.7] record several equivalent Type-II criteria, including a form that they attribute to Rosati. We did not independently consult Rosati’s 1954 paper and therefore do not claim that (9)–(10) occur there in this form. The contribution calculated here is the cutoff, its hard-prime refinement and equality, and the complete disjoint return. A Type-II existence theorem would be stronger than the original two-channel conjecture; no such existence theorem is asserted.

The two prime-exponent boxes remain different objects. Chart (9) starts with $u \mid j^2$; its output is the newly reconstructed integer a and the original box $u \mid a^2$. They share the retained integer u , not a falsely identified prime-factor inventory. In particular, $u \mid j$ is not sufficient as a replacement: the $p = 2521$ example below needs $j = 12, u = 16$.

Finite candidate bounds. There are $K(p)$ possible values of each small modulus. The direct chart tests exactly at most

$$\frac{1}{2} \sum_{j=1}^{K(p)} [\tau_{\text{div}}(((p+4j-1)/4)^2) - 1]$$

original divisors. The cofactor chart tests at most

$$\boxed{\sum_{j \leq K(p)} \tau_{\text{div}}(j^2) \leq K(p) \left(\sum_{d \leq K(p)} \frac{1}{d} \right)^2 \leq K(p)(1 + \log K(p))^2.} \quad (11)$$

Indeed $\tau(j^2) = \sum_{d \mid j} 2^{\omega(d)} \leq d_3(j)$, and $\sum_{j \leq K} d_3(j) = \sum_{abc \leq K} 1 \leq K(\sum_{a \leq K} 1/a)^2$. These are candidate counts, not a claim of polynomial-time factorization.

4. A SMALLER COFACTOR IS NOT A SMALLER ORIGINAL SHELL

At $p = 2521$, the original middle state

$$(a, R, Q, u, h, r, s, \lambda, j) = (644, 55, 47, 16, 1, 4, 161, 3, 12)$$

is returned by the cofactor chart. The original shell at residual 47 has

$$a' = (2521 + 47)/4 = 642 = 2 \cdot 3 \cdot 107.$$

Its complete square-divisor residues modulo 47 are obtained by multiplying

$$\{1, 3, 9, 2, 6, 18, 4, 12, 36\}$$

by 1, 13, 28. None is the middle target 16 or the exterior target 35. Thus both of that shell’s channels are empty. Simply interchanging R, Q and calling the result a state of the other original shell would be false. The valid inverse instead uses $j = 12, u \mid 12^2$ and (10).

An actual tie is $p = 8929$, $(a, u, h, r, s, \lambda) = (2256, 24, 24, 1, 94, 1)$, for which $R = Q = 95$. It is counted once, in the direct chart. All shared prime factors of the two cofactors are retained.

5. NO FIXED CUTOFF ON THE TWO ORIGINAL GRADES

Write $(\cdot/p)$ for the Legendre symbol and define

$$\nu_p = \min\{n \geq 1 : (n/p) = -1\},$$

$$m_p = \min\{m \geq 1 : m \equiv 3 \pmod{4}, (m/p) = -1\}, \quad J_p = (m_p + 1)/4.$$

These minima range over positive integers, not only primes. The exterior grade is the h in the exterior normalization above; the oriented middle grade is $j = hr\lambda$.

Theorem 5.1 (An unconditional bounded-grade obstruction). *For every integer $B \geq 1$, there are infinitely many hard primes p for which no original E state has $h \leq B$ and no oriented M state has $j = hr\lambda \leq B$. These primes nevertheless have explicit endpoint E solutions.*

Proof. Dirichlet’s theorem in the reduced class $3 \pmod{4}$ first supplies a prime $q \equiv 3 \pmod{4}$ with $q > \max(7, 4B)$. Put

$$L = 8 \prod_{l < q \text{ odd prime}} l.$$

The CRT conditions

$$p \equiv 1 \pmod{L}, \quad p \equiv -1 \pmod{q} \quad (12)$$

define one class modulo Lq , since $q \nmid L$. It is reduced: every prime dividing L sees residue 1, while q sees residue -1 . Dirichlet's theorem [3, Theorem 18.1] supplies infinitely many primes in it; discard the finitely many at most q . The class is $1 \pmod{840}$.

Every odd prime $l < q$ is a quadratic residue modulo p , because quadratic reciprocity and $p \equiv 1 \pmod{l}$ give $(l/p) = (p/l) = 1$; also $(2/p) = 1$ since $p \equiv 1 \pmod{8}$ [2, §6.4, especially pp. 207–213]. The prime q is a nonresidue because $(q/p) = (p/q) = (-1/q) = -1$. Every positive integer below q factors over the preceding residue primes. Multiplicativity therefore proves

$$\nu_p = m_p = q.$$

Let an exterior state be given. Its gate implies $\gcd(u, R) = 1$. For an odd prime $l \mid u$, one also has $l \mid a$ and $R \equiv -p \pmod{l}$. Since $R \equiv 3 \pmod{4}$, reciprocity gives, with the two signs retained,

$$(l/R) = (-1)^{(l-1)/2}(R/l) = (-1)^{(l-1)/2}(-1/l)(p/l) = (l/p).$$

If $2 \mid u$, then $2 \mid a$, $R \equiv 7 \pmod{8}$, and $(2/R) = 1 = (2/p)$. Multiplying with the full prime-power exponents yields $(u/R) = (u/p)$. The gate $4u \equiv -1 \pmod{R}$ now gives $(u/R) = (-1/R) = -1$. Since $u = hr^2$, it follows that $(h/p) = -1$ and hence $h \geq \nu_p = q > B$.

For a middle state, every odd prime $l \mid h$ satisfies $Q \equiv -1 \pmod{l}$. Because $Q \equiv 3 \pmod{4}$, Jacobi reciprocity gives $(l/Q) = (-1)^{(l-1)/2}(Q/l) = 1$. If $2 \mid h$, then $Q \equiv 7 \pmod{8}$ and $(2/Q) = 1$. Thus $(h/Q) = 1$. Moreover $\gcd(hr, Q) = 1$ and (2) gives $p \equiv -4hr^2 \pmod{Q}$, whence $(p/Q) = -1$. Since $p \equiv 1 \pmod{4}$, reciprocity introduces no further sign:

$$(Q/p) = (p/Q) = -(h/Q) = -1.$$

Since $Q \equiv 3 \pmod{4}$, $Q \geq m_p = q$, so $j = (Q+1)/4 \geq (q+1)/4 \geq B+1 > B$. Finally (12) gives the original E state

$$R = q, \quad a = u = (p+q)/4, \quad h = a, \quad r = s = 1, \quad \kappa = (p+1)/q,$$

with ordered solution $(a, a\kappa, p\kappa)$. Thus these are genuine solution-bearing primes, not proposed counterexamples. Explicitly, $p \equiv 1 \pmod{4}$, $q \equiv 3 \pmod{4}$ and $p \equiv -1 \pmod{q}$ make a and κ integral; $0 < q < p$ gives $p/4 < a < p/2$; and $R = 4a - p = q$, $u \mid a^2$, $4u + 1 = p + q + 1 \equiv 0 \pmod{q}$. Finally

$$\frac{1}{a} + \frac{1}{a\kappa} + \frac{1}{p\kappa} = \frac{1}{a} + \frac{p+1}{p\kappa} = \frac{1}{a} + \frac{q}{pa} = \frac{4}{p}.$$

□

This theorem concerns the specific fixed grades h and j . It does not exclude p -dependent cutoffs, a cutoff relative to ν_p or J_p , a finite residual criterion, or a different algorithm. Its progression is endpoint-soluble; it is not a claimed obstruction in the endpoint-free remaining branch.

The supplied finite example has a separate partial-factor order certificate. Let $n > 1$, and let the positive integer F be completely factored. If $F \mid n-1$, $F^2 > n$, and for every prime $l \mid F$ there is a_l with

$$a_l^{n-1} \equiv 1 \pmod{n}, \quad \gcd(a_l^{(n-1)/l} - 1, n) = 1,$$

then n is prime. Indeed, if d is any prime divisor of n , reduction modulo d shows that the order of a_l divides both $d-1$ and $n-1$. The gcd condition says that this order does not divide $(n-1)/l$; consequently it contains the full l -primary part of $n-1$. Doing this for every $l \mid F$ proves $F \mid d-1$. Were n composite, its least prime divisor would satisfy $d \leq \sqrt{n} < F$ while $d \equiv 1 \pmod{F}$, a contradiction. A probable-prime screen used to discover the supplied example is not an input to this final order proof.

6. REMAINING UNIVERSAL OBLIGATION

The square-root atlas is complete for M but can still, logically, be empty. It is a terminating decision procedure for a fixed input, not a proof that it returns a witness. E remains a separate original channel. The fixed-grade obstruction explains why a constant h/j inventory cannot close the problem; it is not a substitute for a lower bound on the full source.

Turn 7's requested universal existence, uniformly positive bound or finite *prime-parameter* cutoff has therefore not been discharged. The exact remaining assertion is the positivity of at least one original E/M coefficient for every prescribed hard prime. No theorem here assumes that assertion.

The standard-library replay compares complete first-half M enumeration with both small charts at every $p \equiv 1 \pmod{8}$ prime through its declared bound. A separate checker enumerates the primitive (h, r, s) domain without importing the first program. Every original state, its orientation, the cofactor equality cases and the invalid shell-swap fixture are retained. Execution receipts state actual outcomes

and bounds; no independent mathematical review, formal Lean build, new overall ES verification range or historical-priority conclusion is claimed.

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